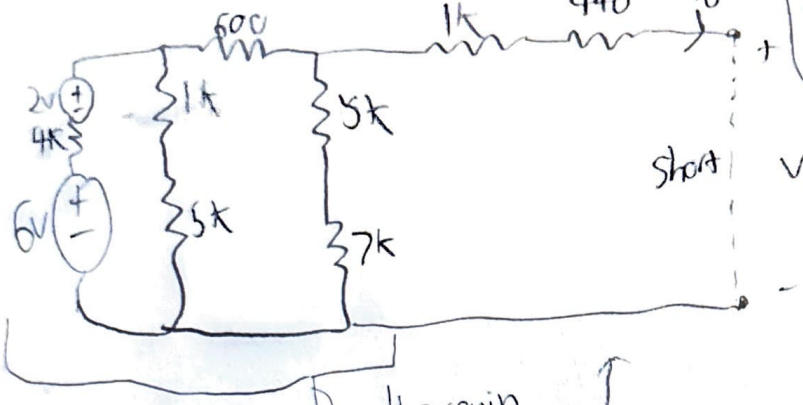
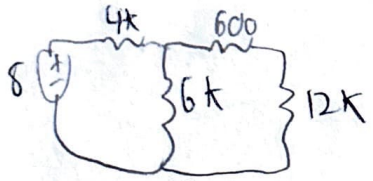
ECEN115 Dylan Thornburg
HW1



short V_0 open

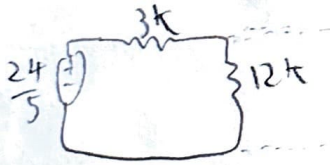
Do thevenin on this section



$Z_{th} = \frac{4 \cdot 6}{4+6} \cdot 1000 = 2.4k$

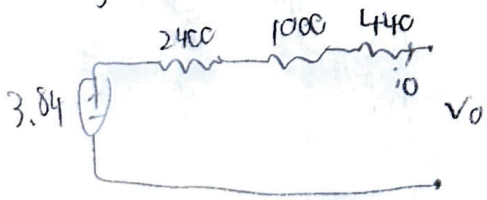
$V_{th} = \frac{6}{6+4} \cdot 8 = \frac{24}{5} V$

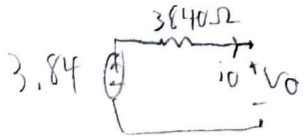
$2.4k + 600 = 3k$



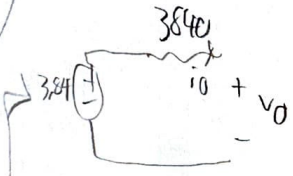
$Z_{th} = \frac{12 \cdot 3}{12+3} = \frac{36}{15} k = 2.4K$

$\frac{24}{5}\left(\frac{12}{3+12}\right) = 3.84V$

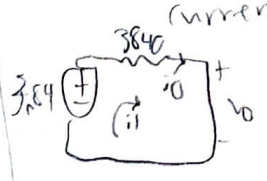




although it looks in series, it is not.



no current, but technically a voltage drop of 3.84 even though there is no current.



$-3.84 + 3840 i_1 = 0$

$3840 i_1 = 3.84$

$i_1 = \frac{3.84}{3840}$

$i_1 = .001$

V_0 is just a wire w/ no change so 0V.

I_0 however = $\frac{3.84}{3840} = .001 A$ or 1mA

open circuit

$i_0 = 0A$

$V_0 = 3.84V$

short circuit

$i_0 = 1mA$

$V_0 = 0V$